Pseudocode

START PasswordManager

PROCESS declare accounts list, choice, isRunning

PROCESS set isRunning to TRUE

IF saved accounts file exists

PROCESS load saved accounts from file into accounts list

ELSE

PROCESS start with an empty accounts list

END IF

WHILE isRunning is TRUE

DISPLAY "===== PASSWORD MANAGER ====="

DISPLAY "1. Add Account"

DISPLAY "2. View All Accounts"

DISPLAY "3. Search Account"

DISPLAY "4. Update Account"

DISPLAY "5. Delete Account"

DISPLAY "6. Generate Random Password"

DISPLAY "7. Exit"

INPUT choice

IF choice is 1

PROCESS run AddAccount

ELSE IF choice is 2

PROCESS run ViewAccounts

ELSE IF choice is 3

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        PROCESS run SearchAccount
        ELSE IF choice is 4
        PROCESS run UpdateAccount
        ELSE IF choice is 5
        PROCESS run DeleteAccount
        ELSE IF choice is 6
        PROCESS run GeneratePassword and get password
        DISPLAY "Generated password: ", password
        ELSE IF choice is 7
        PROCESS save accounts list to file
        DISPLAY "Goodbye!"
        PROCESS set isRunning to FALSE
        ELSE
        DISPLAY "Invalid choice. Please enter a number from 1 to 7."
        END IF

  END WHILE

END PasswordManager




START AddAccount
  INPUT platform
  WHILE platform is empty
        DISPLAY "Platform name cannot be empty."
        INPUT platform
  END WHILE
```

```
INPUT username
WHILE username is empty
        DISPLAY "Username cannot be empty."
        INPUT username
END WHILE

DISPLAY "1. Enter my own password  2. Generate a random password"
INPUT option
WHILE option is not 1 and option is not 2
        DISPLAY "Invalid option. Enter 1 or 2."
        INPUT option
END WHILE

IF option is 2
        PROCESS run GeneratePassword and get password
        DISPLAY "Generated password: ", password
ELSE
        INPUT password
        WHILE password is empty
        DISPLAY "Password cannot be empty."
        INPUT password
        END WHILE
END IF

PROCESS add platform, username, and password to accounts list
DISPLAY "Account saved successfully."
END AddAccount
```

```
START ViewAccounts
  IF accounts list is empty
        DISPLAY "No saved accounts."
  ELSE
        PROCESS set number to 1
        WHILE number is not more than the total accounts
        DISPLAY number, platform, username, and password of that account
        PROCESS add 1 to number
        END WHILE
  END IF
END ViewAccounts

START SearchAccount
  INPUT keyword
  PROCESS set found to FALSE
  PROCESS set number to 1
  WHILE number is not more than the total accounts
        IF platform of that account contains keyword (ignore uppercase and lowercase)
        DISPLAY platform, username, and password of that account
        PROCESS set found to TRUE
        END IF
        PROCESS add 1 to number
  END WHILE
  IF found is FALSE
        DISPLAY "No matching account found."
  END IF
END SearchAccount
```

```
START UpdateAccount
  IF accounts list is empty
        DISPLAY "No accounts to update."
  ELSE
        PROCESS run ViewAccounts
        INPUT accountNumber
        IF accountNumber is not a number or is less than 1 or is more than the total accounts
        DISPLAY "Invalid account number."
        ELSE
        INPUT newUsername
        INPUT newPassword
        IF newUsername is not empty
        PROCESS replace username of the selected account
        END IF
        IF newPassword is not empty
        PROCESS replace password of the selected account
        END IF
        DISPLAY "Account updated successfully."
        END IF
  END IF
END UpdateAccount



START DeleteAccount
  IF accounts list is empty
        DISPLAY "No accounts to delete."
  ELSE
        PROCESS run ViewAccounts
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```
        INPUT accountNumber
        IF accountNumber is not a number or is less than 1 or is more than the total accounts
        DISPLAY "Invalid account number."
        ELSE
        DISPLAY "Are you sure you want to delete this account? (Y/N)"
        PROCESS run ReadYesNo and get confirm
        IF confirm is Y
        PROCESS remove the selected account from accounts list
        DISPLAY "Account deleted successfully."
        ELSE
        DISPLAY "Deletion cancelled."
        END IF
        END IF
  END IF
END DeleteAccount

START GeneratePassword
  DISPLAY "Enter password length (8 to 32):"
  INPUT length
  WHILE length is not a number or is less than 8 or is more than 32
        DISPLAY "Invalid length. Enter a number from 8 to 32."
        INPUT length
  END WHILE

  DISPLAY "Include letters? (Y/N)"
  PROCESS run ReadYesNo and get useLetters
  DISPLAY "Include numbers? (Y/N)"
  PROCESS run ReadYesNo and get useNumbers
  DISPLAY "Include symbols? (Y/N)"
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```
  PROCESS run ReadYesNo and get useSymbols


  IF useLetters is N and useNumbers is N and useSymbols is N
        DISPLAY "Select at least one type. Letters will be used."
        PROCESS set useLetters to Y
  END IF


  PROCESS build character pool from the selected types
  PROCESS set password to empty
  PROCESS set count to 1
  WHILE count is not more than length
        PROCESS pick one random character from character pool
        PROCESS add the character to password
        PROCESS add 1 to count
  END WHILE
END GeneratePassword



START ReadYesNo
  INPUT answer
  WHILE answer is not Y and answer is not N
        DISPLAY "Invalid answer. Enter Y or N."
        INPUT answer
  END WHILE
END ReadYesNo
```